

Letter to the Editor

FRACTIONATED STEREOTACTIC RADIOTHERAPY FOR CHOROIDAL MELANOMA

To the Editor,

In their article, Tokuuye et al. [1] present a technique of irradiation of choroidal melanoma at a 6 MV linear accelerator. A heat-flexible plastic mold was used for head immobilization and 'as the plastic cooled down, the area over the orbit was gently pressed down to restrict ocular movements'. During delineation the patient was only 'instructed to gaze straight forward' [1].

We want to (1) comment generally on eye movements during stereotactic irradiation of ocular targets and (2) present a simple device to overcome this problem. Buatti et al. [2] have already criticized this article on some other technical aspects.

During stereotactic treatment of ocular targets, the overall geometrical inaccuracy of stereotactic positioning is enlarged by the possibility of inadvertent eye movements. Thus, the procedure for defining the reproducibility of the globe position is of utmost importance. Proton radiotherapy of eye tumours usually includes a combination of an active as well as a passive system for proper beam alignment. The patient actively gazes at a given light point. The stability of the gaze position is verified by a TV camera. The passive system in proton beam radiotherapy is based on the relocation of episclerally sutured radioopaque markers. The target repositioning is ensured for every single fraction. The accuracy of this method of target positioning is rarely given. The eye movements during treatment are reported as 0.5 ± 0.3 mm with maximal movements of 1.2 mm. However, interruptions of proton irradiation due to inadvertent eye movements are rare [4].

For Gamma Knife irradiation of eye tumours, passive eye stabilization systems, like bridle sutures through the eye muscles as well as a non-invasive corneal vacuum ring [5], have been reported.

For LINAC treatment of ocular tumours, we have developed a non-invasive active fixation system. This system comprises the use of a fixation light projected via glass fibre cable to the healthy eye. The patient is asked to fixate the fixation light. A mini TV camera is mounted in front of the tumour-containing eye. On a monitor at the control desk the stability of the gaze position is observed and recorded

for analysis. In case of gaze deviation beyond a threshold of 1 mm, the treatment is interrupted. Since the use of CT/MRI image fusion for tumour delineation, additional episcleral markers [6] are rarely necessary. In 10 patients treated with stereotactic photon radiotherapy of uveal melanoma, video analysis with 1164 measurements resulted in a mean horizontal deviation of ± 0.3 mm. The mean vertical deviation was ± 0.2 mm. The maximum deviation never exceeded 1.2 mm.

In our opinion, this active optical fixation system and the possibility of quality control via a TV camera is clearly advantageous to the method described by Tokuuye et al. [1]. At our institution, this system has replaced the formerly used passive vacuum fixation ring system. It is an adequate localization system for treating ocular targets and is clearly superior to the inadequate procedure described by Tokuuye et al. [1].

References

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Sincerely

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